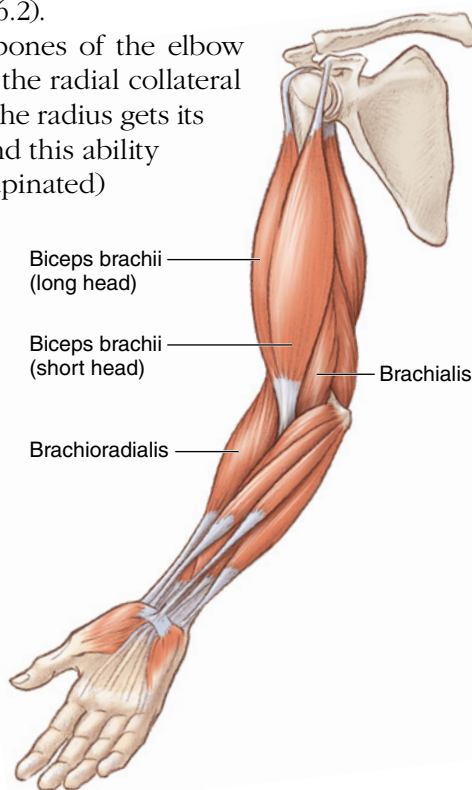


# ARMS, WRISTS, AND HANDS

The major joint of the arm is the elbow and is made up of three bones. The humerus is the long bone in the upper arm and located proximal (closest) to the body, while the radius and ulna are the bones of the forearm and lie distally (farthest from the body). The elbow is a hinge and thus has the capacity only to flex or extend. As a result, the muscles that flex the elbow (biceps brachii, brachialis, brachioradialis, pronator teres) are located anteriorly (on the front; figure 6.1), whereas the extensor muscles (anconeus, triceps brachii) are located posteriorly (on the back; figure 6.2).

The ligaments that help hold the three bones of the elbow joint in place are the joint capsule ligament, the radial collateral ligament, and the ulnar collateral ligament. The radius gets its name from its ability to roll over the ulna, and this ability allows the palm to face either forward (supinated) or backward (pronated) when the arms are hanging at the sides of the body. The head of the radius is connected to the ulna via the annular ligament. Two muscles supinate the forearm (biceps brachii and supinator) so that the palms face forward when the arms hanging down, and two muscles pronate (pronator teres and pronator quadratus) the forearms so the palms face backward.

The degree of available elbow flexion is limited primarily by the forearm contacting the anterior muscles of the upper arm, as well as the anterior proximal ends of the radius and ulna contacting the anterior distal end of the humerus. The tightness of the elbow extensors, however, along with the strength of the elbow flexors and the flexibility of the posterior portions of the capsular, radial collateral, and ulnar collat-



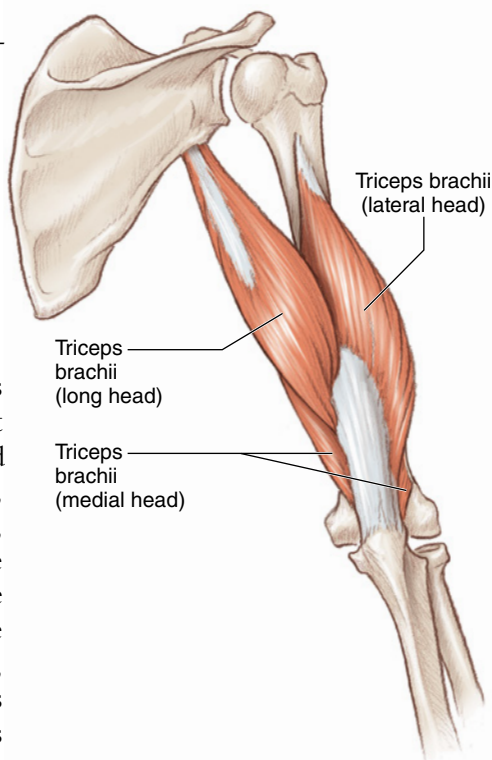
**FIGURE 6.1** Biceps brachii, brachialis, and brachioradialis muscles.

eral ligaments also control the range of movement. These can be altered by stretching.

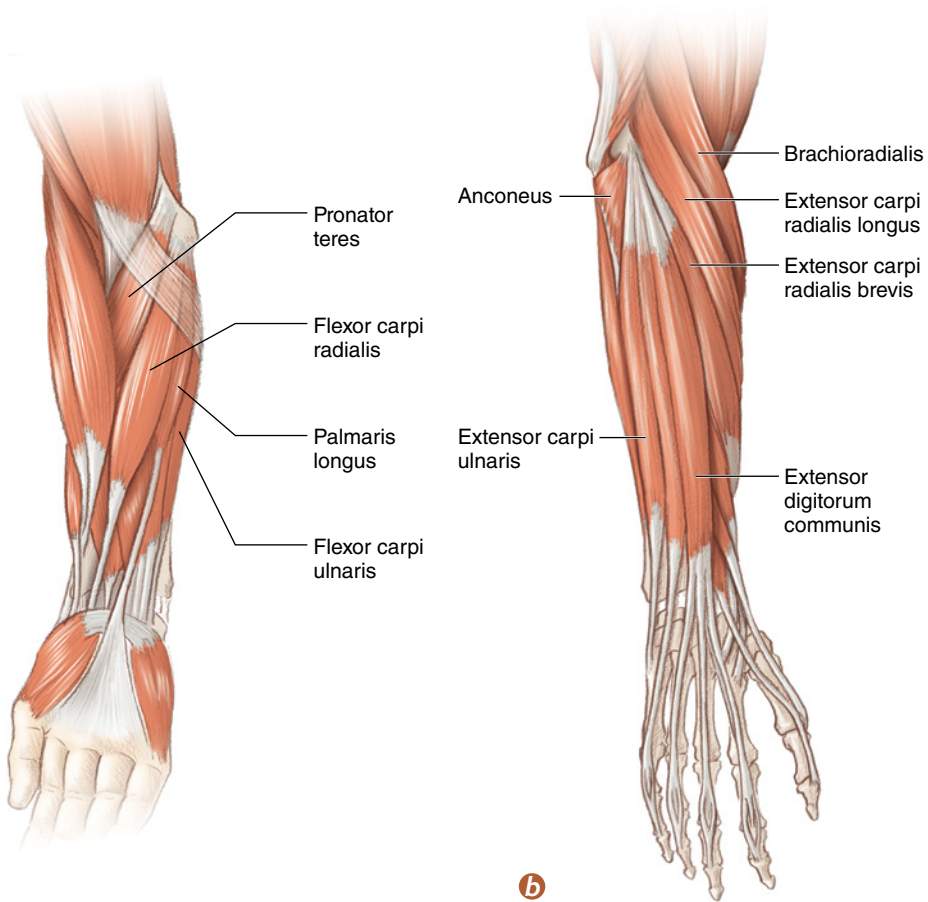
Although the major movements at the wrist are flexion and extension, the wrist is a gliding joint and not a true hinge joint. The gliding is possible because the wrist consists of the distal ends of the radius and ulna and the eight wrist, or carpal, bones. Thus, in addition to flexion and extension, the wrist can perform abduction (radial deviation) and adduction (ulnar deviation). The carpal bones are mostly held together by the various joint capsules, the palmar radiocarpal ligament, and the dorsal radiocarpal ligament. Interestingly, most of the muscles that control wrist, hand, and finger movements are located at or near the elbow. This results in the belly of the muscle lying near the elbow, with tendons crossing the wrist and attaching to the wrist bones (carpals), hand bones (metacarpals), and finger bones (phalanges). Having only tendons in the wrists and hands prevents the wrists and hands from getting too bulky from the increase in size that accompanies muscle strength.

Similar to the muscles that move the elbow, all the wrist flexors (flexor carpi radialis, flexor carpi ulnaris, and palmaris longus) and most of the finger flexors (flexor digitorum profundus, flexor digitorum superficialis, and flexor pollicis longus) are located in the anterior compartment of the forearm (figure 6.3a). In contrast, all the wrist extensors (extensor carpi radialis brevis, extensor carpi radialis longus, extensor carpi ulnaris, extensor digitorum communis) and finger extensors (extensor digitorum communis, extensor digiti minimi, extensor indicis) are located in the posterior compartment of the forearm (figure 6.3b). The muscles that run along the radius, which have *radialis* in their names, perform ulnar deviation, or wrist abduction. Those along the ulna, which have *ulnaris* in their names, perform radial deviation, or wrist adduction. Just before crossing the wrist, the tendons of these muscles are anchored firmly by thick tissue bands called the flexor retinaculum and extensor retinaculum. By passing under the retinaculum at the carpals (wrist bones), the tendons lie in a carpal tunnel. Because the tendons are crowded together, each tendon is surrounded by a slippery sheath to minimize friction.

The movement ranges for wrist flexion, wrist extension, radial deviation, and ulnar deviation are all limited by the strength of the agonist muscles, flexibility of the antagonist muscles, tightness of the dorsal and palmar ligaments, and wrist impingement (ulnar deviation only). Interestingly, all of these, except wrist impingement, can be changed by doing stretching exercises.



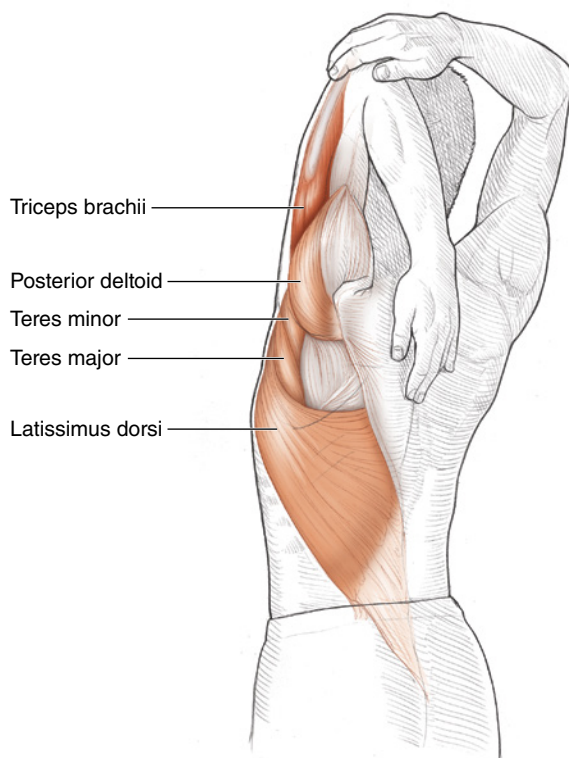
**FIGURE 6.2** Triceps brachii muscle.



**FIGURE 6.3** Forearm muscles: (a) inside; (b) outside.

Stretching the muscles that move the elbows and wrists is helpful in alleviating and sometimes preventing overuse injuries. Because it is more resistive to opposing movements, a tight muscle is easy to damage. When the wrist extensor muscles are tight, pain arises on the lateral (outer) side of the elbow. In sports, this pain is sometimes referred to as tennis elbow. Tight wrist flexor muscles, on the other hand, can cause pain on the opposite, or medial, side of the elbow. This pain is frequently called golfer's elbow. Also, tightness in both the wrist extensors and flexors from either constant wrist hyperextension or flexion can lead to increased friction, inflammation, and overuse injuries such as carpal tunnel syndrome. People engaged in static or fine motor work such as keyboard use, computer mouse use, carpentry, or rock climbing are most likely to encounter this condition. To prevent and alleviate this condition, rehabilitation specialists encourage work breaks for stretching both the wrist flexors and extensors to help strengthen and loosen the muscles and tendons.

# TRICEPS BRACHII STRETCH



## Execution

1. Sit in a chair with a back or stand upright with the left arm flexed at the elbow.
2. Raise the left arm until the elbow is next to the left ear and the left hand is near the right shoulder blade.
3. Grasp the upper arm just below the left elbow with the right hand, and pull or push the left elbow behind the head and toward the floor.
4. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscle:** Left triceps brachii

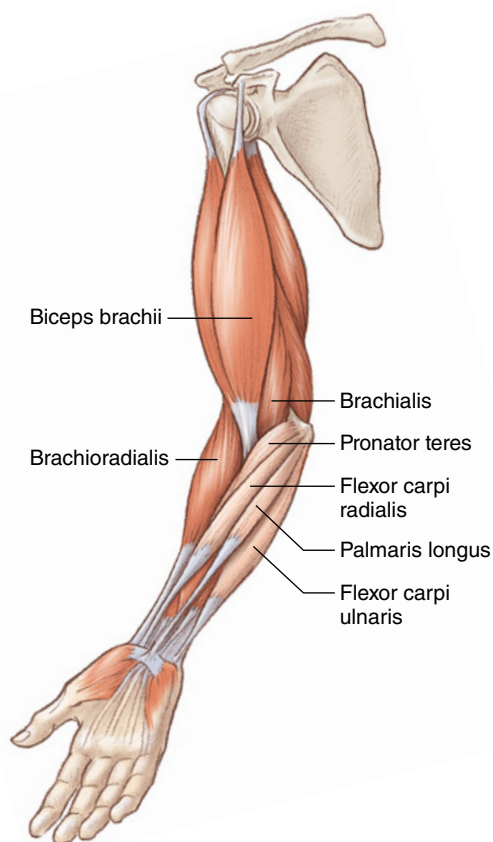
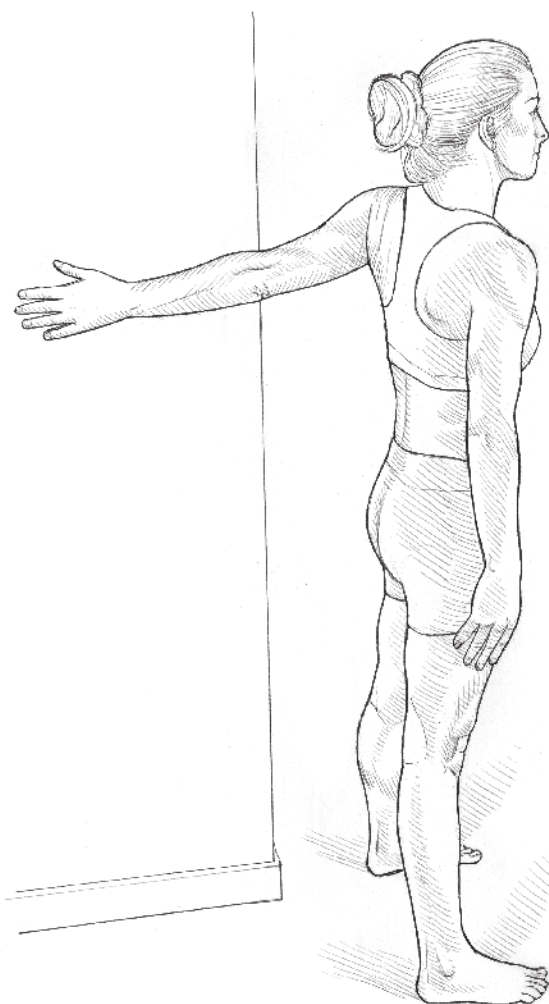
**Less-stretched muscles:** Left latissimus dorsi, left teres major, left teres minor, left posterior deltoid

## Stretch Notes

Tightness in the elbow extensor muscles is the main cause of tennis elbow, or pain in the lateral elbow during arm movements. This tightness is usually caused by overworking or straining these muscles or by working against resistance with the arm fully extended. Therefore, any activity that uses these muscles can lead to tightness. Consequently, this stretch is beneficial not only for tennis players but also for swimmers. Alternatively, strain can result if the muscle is constantly overstretched by tight elbow flexors or if the arm is muscle bound (inability to completely straighten the arm).

Doing this stretch while seated in a chair with a back allows better control of balance. A greater stretching force can be applied to the muscles when the body is balanced. Also, do not perform this stretch for an extended period because this stretch greatly reduces blood flow to the shoulder.

## ELBOW FLEXOR STRETCH



### Execution

1. Stand facing the inside of a doorframe, at an arm's length.
2. Raise the left arm to shoulder level, keeping it straight.
3. Grasp the farthest edge of the doorframe or lay the left hand flat against the wall next to the doorway, with the thumb pointing up.
4. Keeping the left elbow and wrist straight, rotate the trunk back toward the doorframe or doorway.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left brachialis, left brachioradialis, left biceps brachii

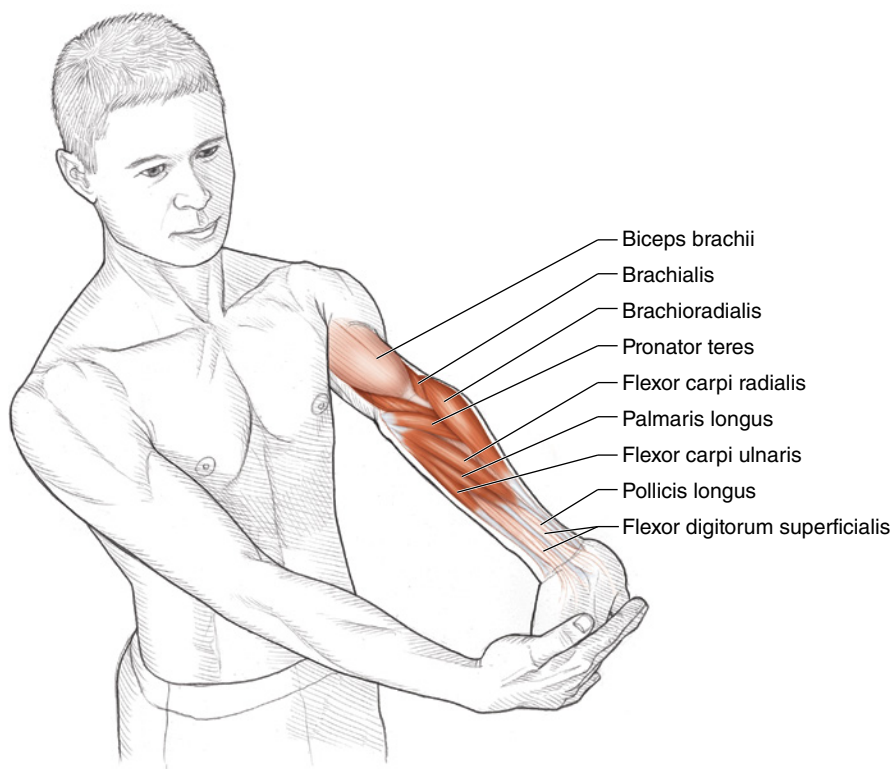
**Less-stretched muscles:** Left supinator, left pronator teres, left flexor carpi radialis, left flexor carpi ulnaris, left palmaris longus

## Stretch Notes

These flexor muscles easily become tight from large amounts of bent-elbow work such as carrying heavy boxes or curling either dumbbells or barbells. When these muscles are tight, the arm cannot be completely straightened, and the person has what is often called a muscle-bound look. This tightness causes pain on the medial elbow, often referred to as golfer's elbow. However, the pain is not limited to golfers and can affect other people such as carpenters, rock climbers, massage therapists, and weightlifters. Also, stretching these flexor muscles can bring relief to those who experience carpal tunnel syndrome.

This stretch is easier to do if you grasp a solidly fixed vertical pole. Grasp the pole firmly so your hand does not slide along the pole, but do not grasp too tightly because a tight grasp virtually eliminates the stretch effect on the less-stretched muscles. Also, it is more difficult to keep the elbow straight, and a straight elbow is necessary for this stretch to be effective. It is preferable to lift the arm to shoulder level to ensure that all muscles receive the same amount of stretch. Nevertheless, the stretch will be effective at whatever height the arm is raised.

## ELBOW AND WRIST FLEXOR STRETCH



### Execution

1. Stand upright with feet shoulder-width apart, toes pointing straight forward.
2. Stick your left arm out in front of you at shoulder height, with the elbow straight and forearm supinated (turned up).
3. Hyperextend the left wrist so that the fingers point toward the floor.
4. Grab the left fingers with the right hand, and pull the fingers back toward the elbow.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left brachialis, left brachioradialis, left pronator teres, left flexor carpi radialis, left flexor carpi ulnaris, left palmaris longus

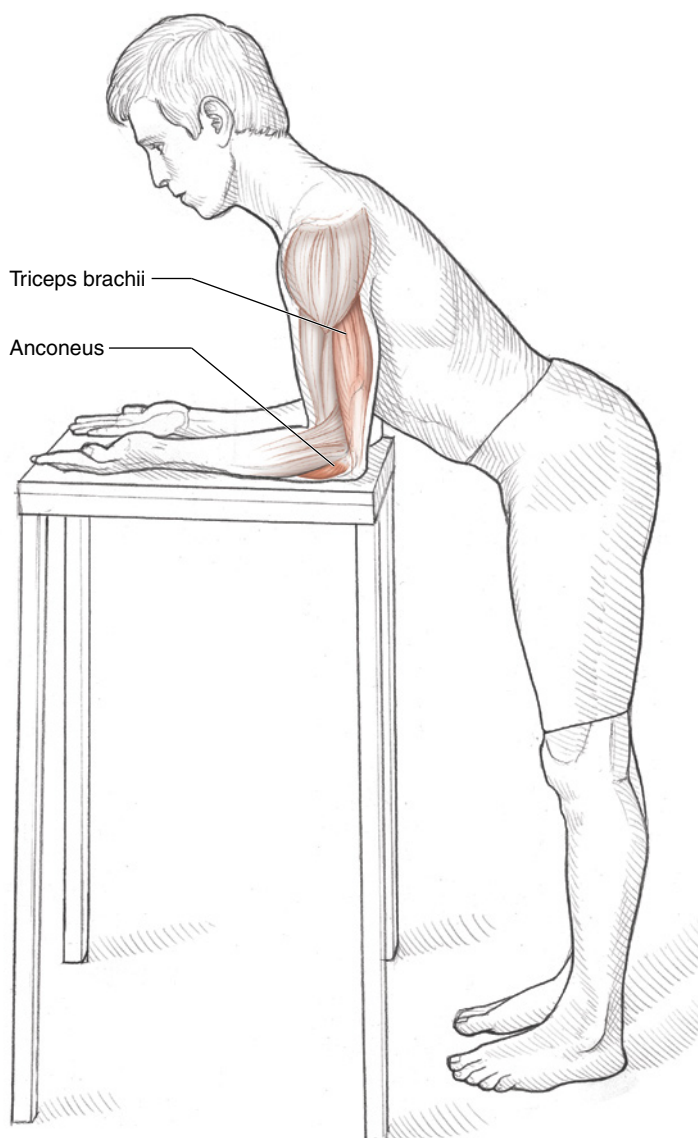
**Less-stretched muscles:** Left biceps brachii, left flexor digitorum superficialis, left flexor digitorum profundus, left pollicis longus

## Stretch Notes

These flexor muscles easily become tight from static work such as operating a keyboard. Also any occupation that requires large amounts of arm work can cause these muscles to become tight. This tightness causes pain on the medial elbow, often referred to as golfer's elbow. However, the pain is not limited to golfers and can affect other people such as carpenters, rock climbers, and massage therapists. Also, stretching these flexor muscles can help bring relief to those with carpal tunnel syndrome.

Exercise caution when doing this stretch. If you feel pain in the elbow, wrist, or finger joints, reduce the tension or joint damage could occur.

## ANCONEUS STRETCH



## Execution

1. Stand or sit upright while facing a table that is about waist high.
2. Flex the elbows and rest the forearms on the table with the palms up.
3. Lean forward, bringing the chest toward the table.

## Muscles Stretched

**Most-stretched muscle:** Anconeus

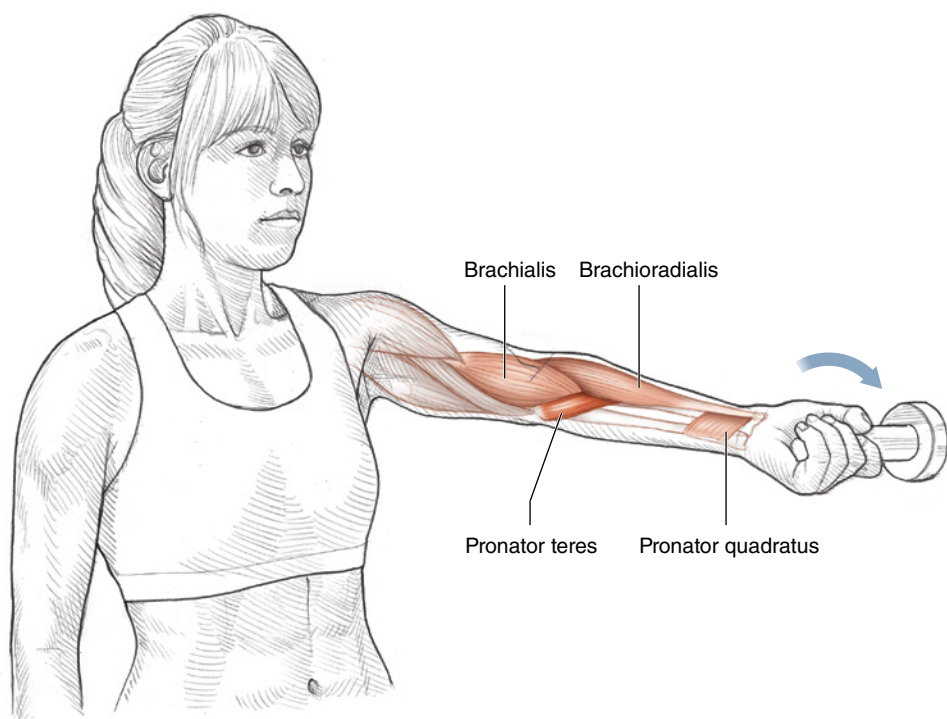
**Less-stretched muscle:** Triceps brachii

## Stretch Notes

Tightness in the elbow extensor muscles is the main cause of tennis elbow, or pain in the lateral elbow during arm movements. This tightness usually is caused by overworking or straining these muscles. Therefore any activity that uses these muscles can lead to tightness. Although the triceps brachii is the major muscle used in extending the elbow, the anconeus becomes a major player when the arm is bent and pronated. Hence, tennis players who mainly use a close-to-the-body forehand stroke or someone who has the muscle-bound look (unable to straighten the arms) will benefit greatly from this stretch.

For the greatest stretch, keep the forearms and elbows flat on the table.

## FOREARM PRONATOR STRETCH WITH DUMBBELL



### Execution

1. Stand upright with the feet shoulder-width apart, toes pointing straight forward.
2. With the left hand, grasp a light dumbbell that has a weight plate attached to one end only, with the weighted end sticking out past the thumb.
3. Hold your left arm out in front of you at shoulder height, with the elbow straight and forearm supinated (with the top of the weight to the left of the thumb).
4. Hypersupinate the forearm (rotate the wrist toward the thumb) so that the weighted end of the dumbbell points toward the floor.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscle:** Left pronator teres

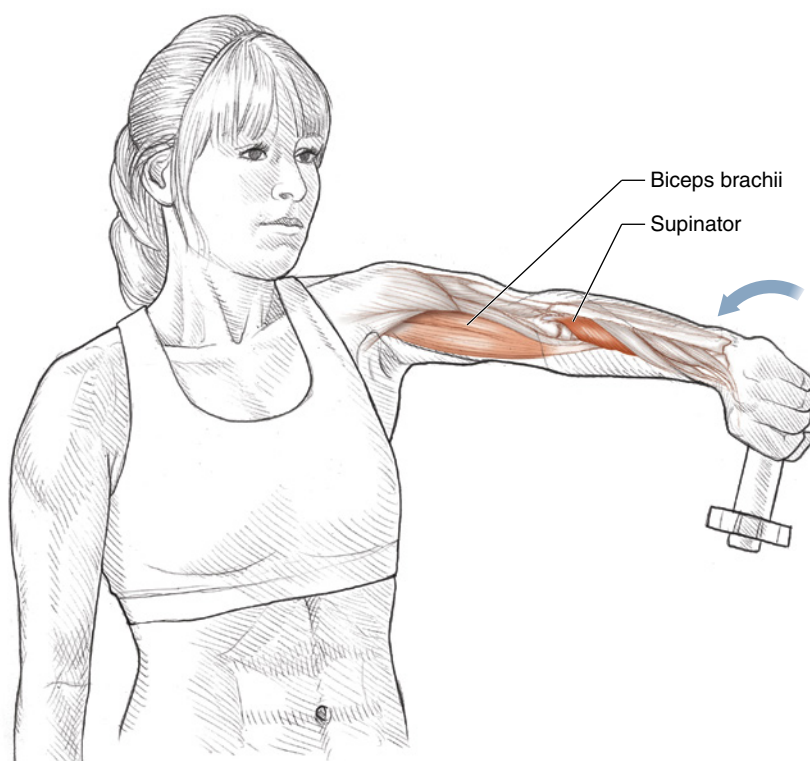
**Less-stretched muscles:** Left brachialis, left brachioradialis, left pronator quadratus

## Stretch Notes

A pronation contracture, or extremely tight pronator muscles, is primarily caused by hypertonicity (a shortened, stiff muscle) in the pronator teres. This hypertonicity can cause medial nerve compression, or pronator teres syndrome. The symptoms are felt as pain and weakness in the anterior forearm and hand. Pronator teres syndrome results from overuse of the pronator teres through repetitive occupational activities such as hammering, cleaning fish, or performing any activity that requires continual manipulation of tools. Women are affected more than men, although the reason for this is not clear. Regularly stretching the pronator teres can help reduce the possibility of developing contractures.

Don't use a weight that is too heavy. Start with a very light weight plate on one end of the dumbbell, and gradually increase the weight as you become more used to the stretch. In fact, you do not need to use a dumbbell at all. Any object that has a light weight on one end of a handle, such as a hammer, will work just as well. Also, this stretch can be done either sitting or standing, with the whole arm lying on a flat surface and the wrist and hand extended past the edge of the surface. If you do use a support, try to keep the shoulder angle near 90 degrees.

## FOREARM SUPINATOR STRETCH WITH DUMBBELL



### Execution

1. Stand upright with your feet shoulder-width apart, toes pointing straight forward.
2. In your left hand, grasp a light dumbbell that has a weight plate attached to one end only, with the weighted end sticking out past the thumb.
3. Hold your left arm out in front of you at shoulder height, with the elbow straight and forearm supinated.
4. Pronate the forearm (rotate the wrist toward the little finger) so that the weighted end of the dumbbell points toward the floor.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscle:** Left supinator

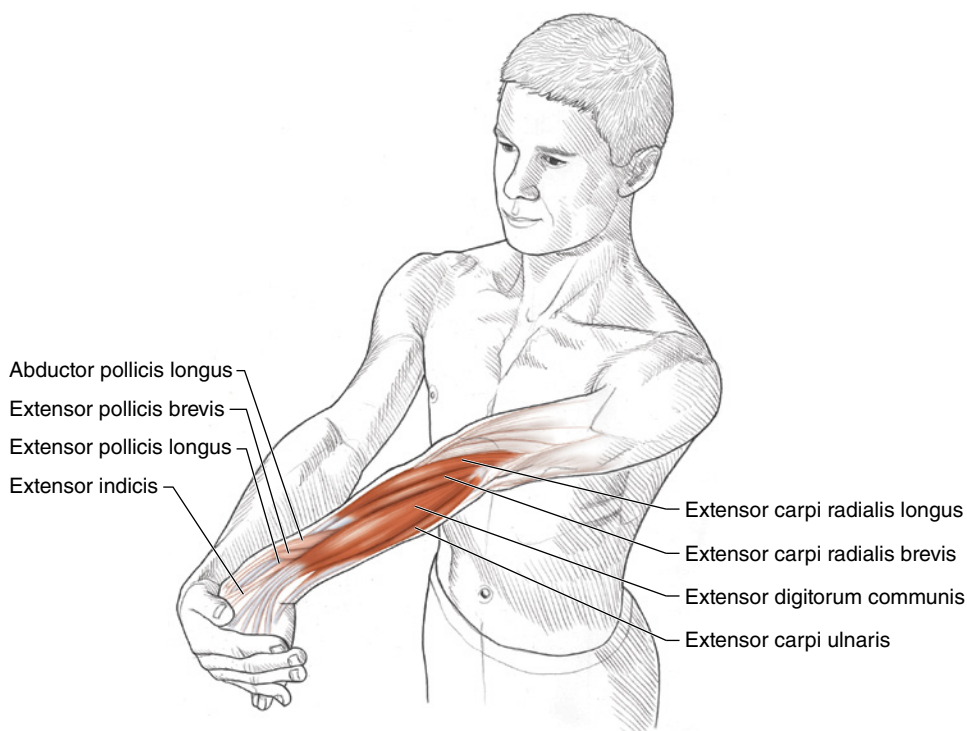
**Less-stretched muscle:** Left biceps brachii

## Stretch Notes

A short and tight (hypertonic) supinator is a major contributor to lateral elbow pain, often called tennis elbow. A severe hypertonic supinator can contribute to either supinator syndrome or radial tunnel syndrome. These syndromes are the result of radial nerve compression and manifest themselves as forearm pain and numbness along with weakness in the lower arm and hand muscles. Movements such as a quick tennis backhand or prolonged forearm supination with a flexed elbow, such as cutting hair, walking a dog on a leash, or carrying heavy boxes from underneath, are the types of movements that can overwork the supinator and lead to a hypertonic muscle.

Don't use a weight that is too heavy. Start with a very light weight plate on one end of the dumbbell, and gradually increase the weight as you become more used to the stretch. In fact, you do not need to use a dumbbell at all. Any object that has a weight on one end of a handle, such as a hammer, will work just as well. Also, this stretch can be done either sitting or standing, with the whole arm lying on a flat surface and the wrist and hand extended past the edge of the surface. If you do use a support, try to keep the shoulder angle near 90 degrees.

# BEGINNER WRIST EXTENSOR STRETCH



## Execution

1. Stand upright with feet shoulder-width apart, toes pointing straight forward.
2. Stick your left arm out in front of you at shoulder height, with the elbow straight and forearm pronated.
3. Bend the left wrist so that the fingers point toward the floor.
4. Place the palm of the right hand against the knuckles of the left hand.
5. While keeping the left elbow straight, pull the knuckles toward the body.
6. Repeat these steps for the opposite arm.

## Muscles Stretched

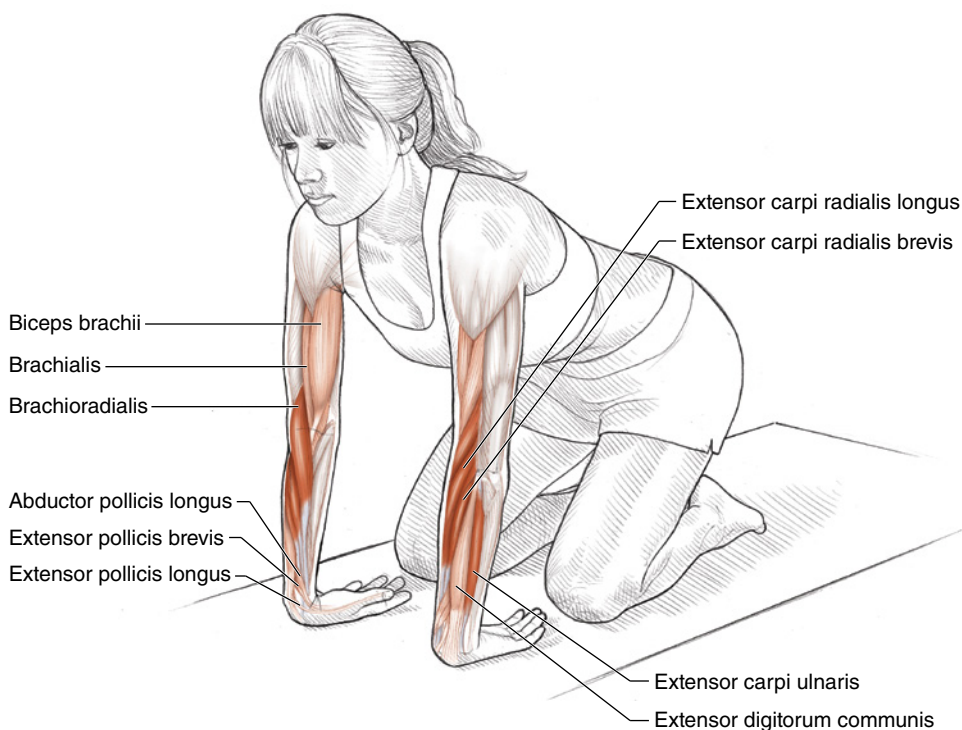
**Most-stretched muscles:** Left extensor carpi radialis brevis, left extensor carpi radialis longus, left extensor carpi ulnaris, left extensor digitorum communis

**Less-stretched muscles:** Left extensor indicis, left extensor pollicis brevis, left extensor pollicis longus, left abductor pollicis longus

## Stretch Notes

Tightness in the extensor muscles is a cause of tennis elbow, or pain in the lateral elbow during arm movements. This tightness usually is caused by overworking or straining these muscles. Any activity that uses these muscles, such as keyboard work, racket sports, rowing, weightlifting, wheelchair sports, and rock climbing, can lead to overwork, hypertonicity, and tightness. Also, overworking the extensor pollicis longus and brevis or the abductor pollicis longus can lead to conditions known as drummer boy palsy (mainly the extensor pollicis longus), intersection syndrome (mainly the extensor pollicis brevis and abductor pollicis longus), and de Quervain syndrome (mainly extensor pollicis brevis and abductor pollicis longus tendons). By doing this stretch, you help reduce the problems that can arise from overworked wrist extensors.

# INTERMEDIATE WRIST EXTENSOR STRETCH



## Execution

1. Kneel on the floor.
2. Flex both wrists and place the backs of your hands on the floor, hands shoulder-width apart.
3. Point the fingers toward the knees.
4. While keeping the elbows straight, lean back, bringing the buttocks to the heels, keeping the backs of the hands on the floor.

## Muscles Stretched

**Most-stretched muscles:** Brachioradialis, extensor carpi radialis brevis, extensor carpi radialis longus, extensor carpi ulnaris

**Less-stretched muscles:** Supinator, brachialis, biceps brachii, extensor digitorum communis, extensor pollicis brevis, extensor pollicis longus, abductor pollicis longus

## Stretch Notes

Tightness in the extensor muscles can cause tennis elbow, or pain in the lateral elbow during arm movements. This tightness usually is caused by overworking or straining these muscles. Therefore, any activity that uses these muscles, such as

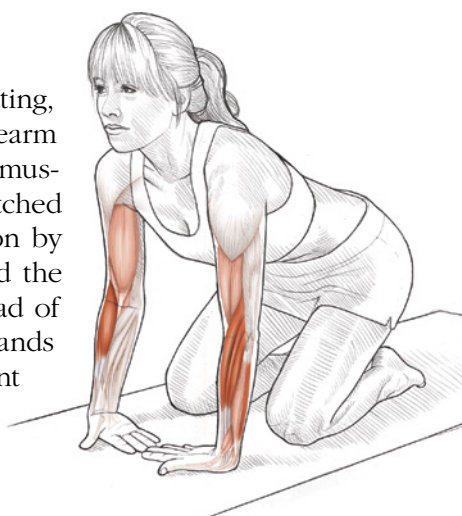
keyboard work, racket sports, rowing, weightlifting, wheelchair sports, and rock climbing, can lead to overwork, hypertonicity, and tightness. Overworking the extensor pollicis longus and brevis or the abductor pollicis longus can lead to conditions known as drummer boy palsy (mainly the extensor pollicis longus), intersection syndrome (mainly the extensor pollicis brevis and abductor pollicis longus), and de Quervain syndrome (mainly the extensor pollicis brevis and abductor pollicis longus tendons). The beginner stretch is best for those who have a small range of wrist motion or severe pain when using the wrist. Once you gain more range of motion, however, you should do this intermediate stretch to reduce the problems that can arise from overworked wrist extensors. This stretch will also strengthen the afflicted muscles and start you on the road to prevention of further problems.

The closer the hands are to the knees, the easier it is to keep the backs of the hands touching the floor. The farther the hands are in front of the knees, however, the greater the applied stretch.

### VARIATION

#### ***Wrist Radial Deviator and Extensor Stretch***

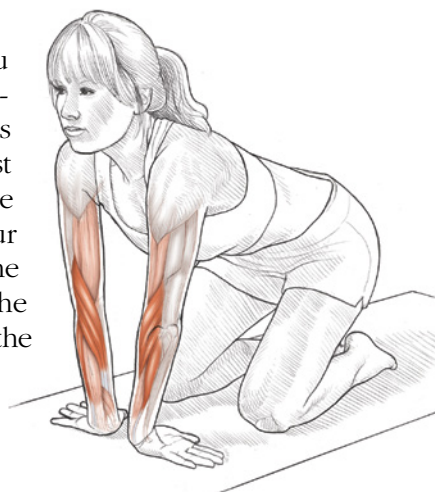
By changing the direction that the fingers are pointing, you can change the stretching emphasis on the forearm muscles. For instance, both the wrist extensor muscles and the radial deviator muscles can be stretched simultaneously. First, assume the starting position by kneeling on the floor with the wrists flexed and the backs of your hands on the floor. Second, instead of pointing the fingers toward the knees, rotate the hands so that the fingers point medially (fingertips point toward each other). Finally, stretch the desired muscles by leaning back (buttocks to the heels) while keeping the backs of the hands on the floor.



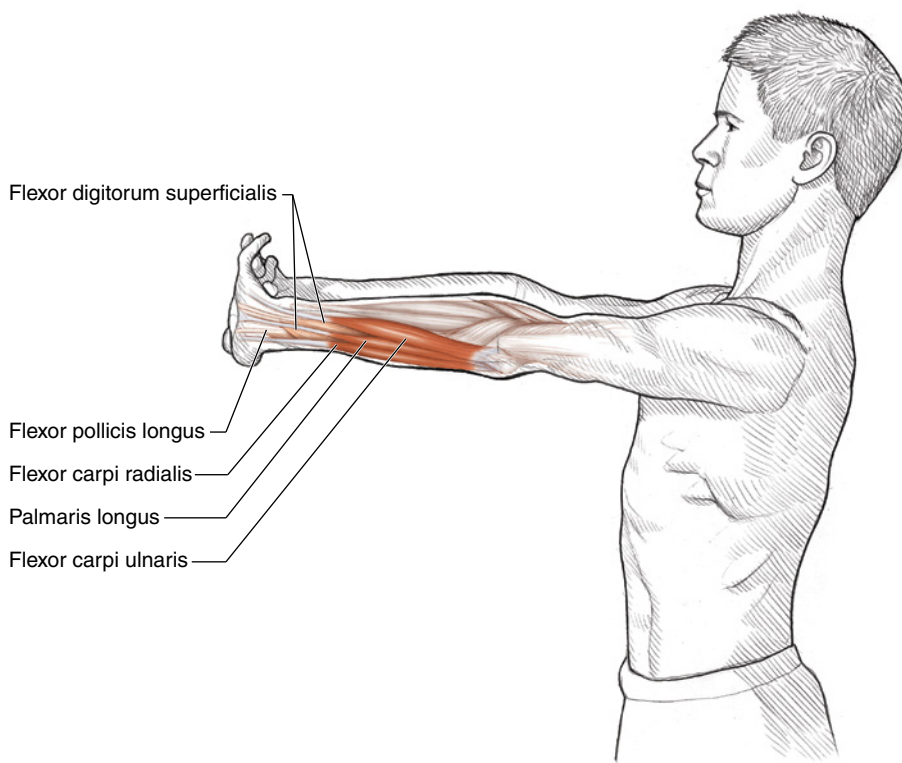
### VARIATION

#### ***Wrist Ulnar Deviator and Extensor Stretch***

If you change the direction the fingers point, you alter the stretching emphasis on the forearm muscles. To stretch both the wrist extensor muscles and the ulnar deviator muscles simultaneously, first assume the starting position by kneeling on the floor with the wrists flexed and the backs of your hands on the floor. Second, instead of pointing the fingers toward the knees, rotate the hands so the fingers point laterally (fingertips point away from the body on a line perpendicular to the midline of the body). Finally, stretch the desired muscles by leaning back (buttocks to the heels) while keeping the backs of the hands on the floor.



# BEGINNER WRIST FLEXOR STRETCH



## Execution

1. Stand upright with feet shoulder-width apart, toes pointing straight forward.
2. Interlock your fingers, with the palms pointing out away from the body.
3. With the arms at shoulder level, straighten your elbows and push your palms out away from the body as far as you can.

## Muscles Stretched

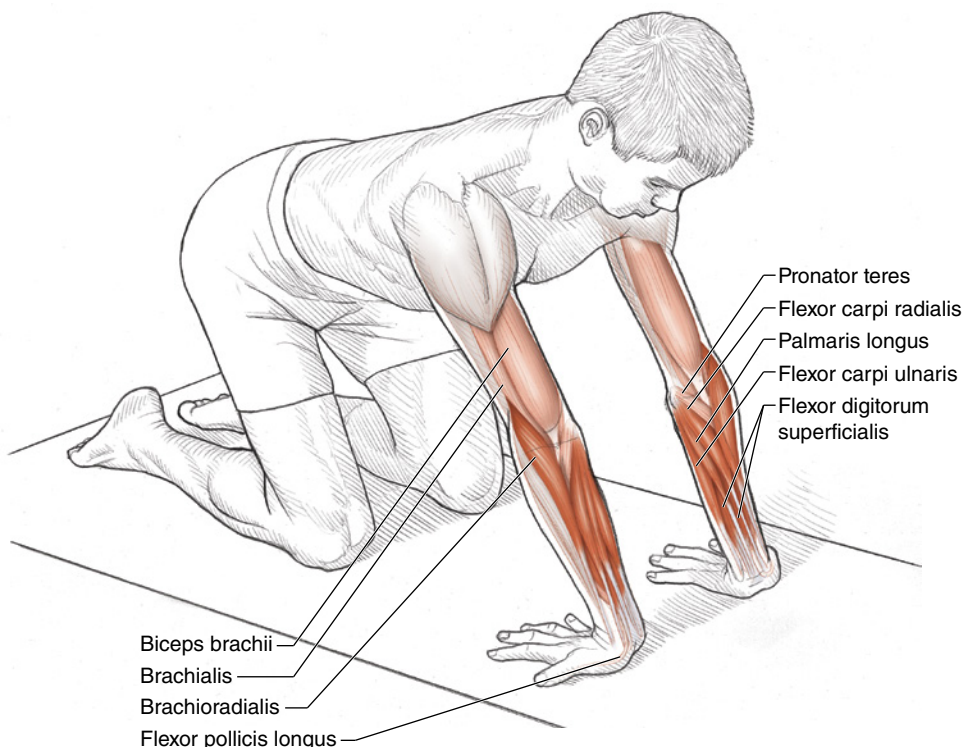
**Most-stretched muscles:** Flexor carpi radialis, flexor carpi ulnaris, pronator teres, palmaris longus

**Less-stretched muscles:** Flexor pollicis longus, flexor digitorum profundus, flexor digitorum superficialis

## Stretch Notes

The flexor muscles easily become tight from repeated use of the arm or wrist in an awkward position or by bending the wrist while typing, using the phone, or operating a machine. Additional problems arise from working with the arm held away from the body or playing sports. This tightness causes pain on the medial elbow, often referred to as golfer's elbow. The longer a person does any of these activities, the greater the risk of tightness and the greater the need to stretch these muscles.

# INTERMEDIATE WRIST FLEXOR STRETCH



## Execution

1. Kneel on the floor.
2. Flex both wrists and place the palms of your hands on the floor, hands shoulder-width apart.
3. Point the fingers toward the knees.
4. While keeping the elbows straight, lean back (buttocks to the heels), keeping the palms flat on the floor.

## Muscles Stretched

**Most-stretched muscles:** Brachioradialis, flexor carpi radialis, flexor carpi ulnaris, flexor digitorum profundus, flexor digitorum superficialis, palmaris longus

**Less-stretched muscles:** Flexor digiti minimi brevis, flexor pollicis longus, pronator teres, brachialis, biceps brachii

## Stretch Notes

The flexor muscles easily become tight from repeated use of the arm or wrist in an awkward position or by bending the wrist while typing, using the phone, or

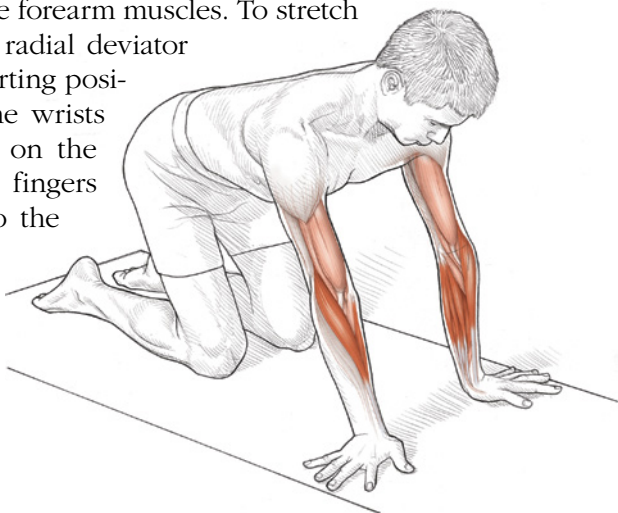
operating a machine. Additional problems arise from working with the arm held away from the body or playing sports. This tightness causes pain on the medial elbow, often referred to as golfer's elbow. The longer a person does any of these activities, the greater the risk of tightness and the greater the need for stretching these muscles. Unfortunately, the beginner-level exercise provides only limited stretch. As you increase your flexibility, you need to move to a more intense stretch, such as this intermediate one.

The closer the hands are to the knees, the easier it is to keep the palms of the hands touching the floor. The farther the hands are away from the midline, the greater the stretch.

### **VARIATION**

#### ***Wrist Radial Deviator and Flexor Stretch***

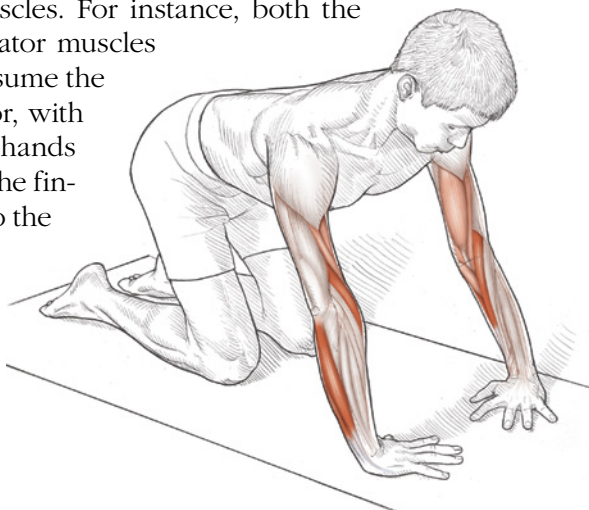
If you change the direction that the fingers are pointing, you alter the stretching emphasis being placed on the forearm muscles. To stretch both the wrist flexor muscles and the radial deviator muscles simultaneously, assume the starting position by kneeling on the floor, with the wrists flexed and the palms of your hands on the floor. Second, instead of pointing the fingers toward the knees, rotate the hands so the fingers point laterally (fingertips point away from the body on a line perpendicular to the midline of the body). Finally, stretch the desired muscles by leaning back (buttocks to the heels) while keeping the palms of the hands on the floor.



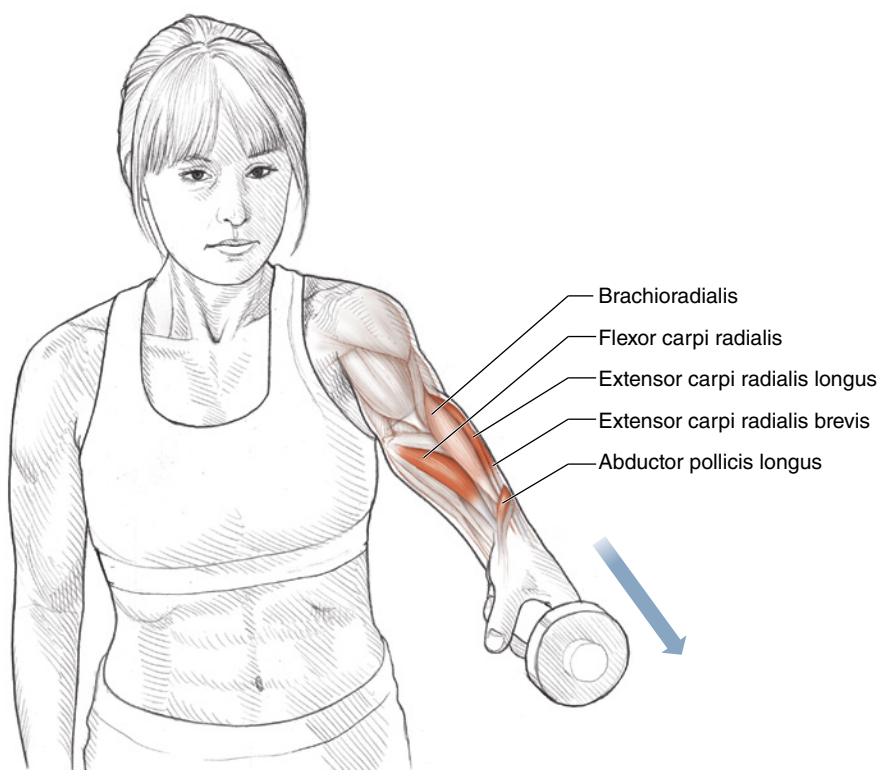
### **VARIATION**

#### ***Wrist Ulnar Deviator and Flexor Stretch***

By changing the direction the fingers are pointing, you change the stretching emphasis on the forearm muscles. For instance, both the wrist flexor muscles and the ulnar deviator muscles can be stretched simultaneously. First, assume the starting position by kneeling on the floor, with the wrists flexed and the palms of your hands on the floor. Second, instead of pointing the fingers toward the knees, rotate the hands so the fingers point medially (fingertips point toward each other). Finally, stretch the desired muscles by leaning back (buttocks to the heels) while keeping the palms of the hands on the floor.



## WRIST RADIAL DEVIATOR STRETCH WITH DUMBBELL



### Execution

1. Stand upright with the feet shoulder-width apart, toes pointing straight forward.
2. In your left hand, grasp a dumbbell that has a weight plate attached to one end only, with the weighted end sticking out past the thumb.
3. Hold your left arm out in front of you at shoulder height, with the elbow straight and forearm rotated so that the thumb side of the hand points up.
4. Bend the wrist down so that the weighted end of the dumbbell points more forward, away from the body rather than up.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left abductor pollicis longus, left flexor carpi radialis, left extensor carpi radialis longus, left extensor carpi radialis brevis

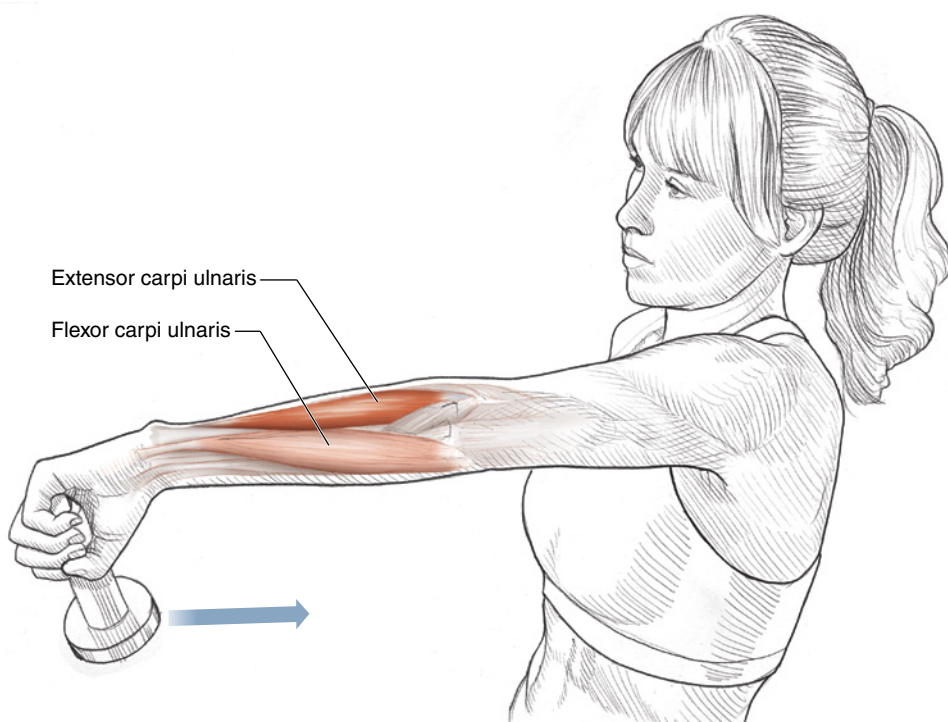
**Less-stretched muscle:** Left brachioradialis

## Stretch Notes

Many activities that require using the wrist in repetitive actions for many hours each day, such as extended work on a computer or tennis, golf, baseball, bowling, and mountain biking, force the wrist joint to the extremes of its range of motion and make the area vulnerable to tightness or hypertonicity. If done without adequate rest and recovery, the limited, repetitive motions involved in playing the violin or piano can also cause tightness. Also, the wrist may be damaged in simple, mundane activities such as scrubbing a pot, pushing up out of a chair, or lifting a small object in an awkward position. Much of the tightness, pain, and injury associated with these activities can be relieved through stretching the wrist radial deviators.

Don't use a weight that is too heavy. Start with a very light weight plate on one end of the dumbbell, and gradually increase the weight as you become more used to the stretch. In fact, you do not need to use a dumbbell at all. Any object with a weight attached to one end of a handle, such as a hammer, will work just as well. Also, this stretch can be done either sitting or standing, with the whole arm lying on a flat surface and the wrist and hand extended past the edge of the surface. If you do use a support, try to keep the shoulder angle near 90 degrees.

## WRIST ULNAR DEVIATOR STRETCH WITH DUMBBELL



### Execution

1. Stand upright with the feet shoulder-width apart, toes pointing straight forward.
2. In the left hand, grasp a dumbbell that has a weight plate attached to one end only, with the weighted end sticking out past the thumb.
3. Hold your left arm out in front of you at shoulder height, with the elbow straight and forearm rotated so that the thumb side of the hand points down.
4. Bend the wrist down so that the weighted end of the dumbbell points more toward the body rather than down.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscle:** Left extensor carpi ulnaris

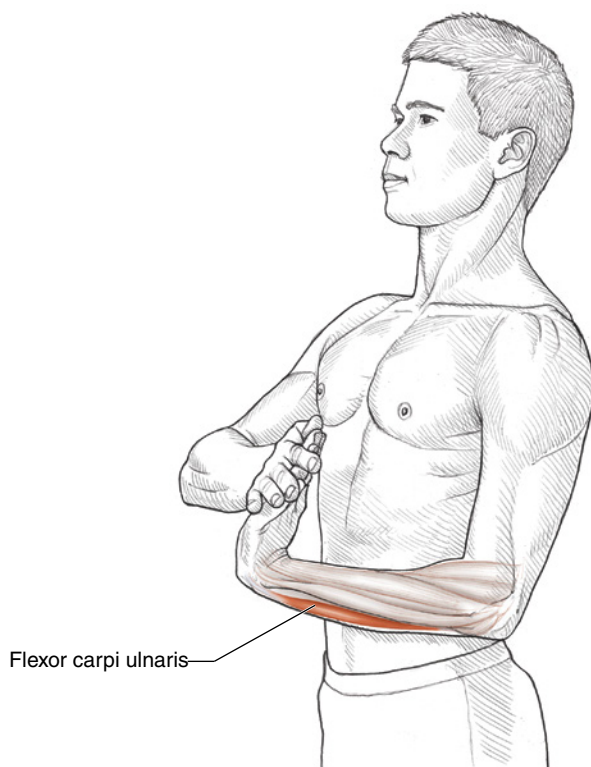
**Less-stretched muscle:** Left flexor carpi ulnaris

## Stretch Notes

Many activities that require using the wrist in repetitive actions for many hours each day, such as extended work on a computer or tennis, golf, baseball, bowling, and mountain biking, force the wrist joint to the extremes of its range of motion and make the area vulnerable to tightness or hypertonicity. If done without adequate rest and recovery, the limited, repetitive motions involved in playing the violin or piano can also cause tightness. Also, the wrist may be damaged in simple, mundane activities such as scrubbing a pot, pushing up out of a chair, or lifting a small object in an awkward position. Much of the tightness, pain, and injury associated with these activities can be relieved through stretching the wrist ulnar deviators.

Do not use a weight that is too heavy. Start with a very light weight plate on one end of the dumbbell, and gradually increase the weight as you become more used to the stretch. In fact, you do not need to use a dumbbell at all. Any object that has a weight attached to one end of a handle, such as a hammer, will work just as well. Also, this stretch can be done either sitting or standing, with the whole arm lying on a flat surface and the wrist and hand extended past the edge of the surface. If you do use a support, try to keep the shoulder angle near 90 degrees.

# FINGER FLEXOR STRETCH



Flexor carpi ulnaris

## Execution

1. Sit or stand upright.
2. Flex the elbow to a 90-degree angle, and extend the wrist as far as possible.
3. Point the fingers upward.
4. With the right hand, push the fingers on the left hand toward the elbow.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left flexor carpi radialis, left flexor carpi ulnaris, left flexor digiti minimi brevis, left flexor digitorum profundus, left flexor digitorum superficialis, left palmaris longus

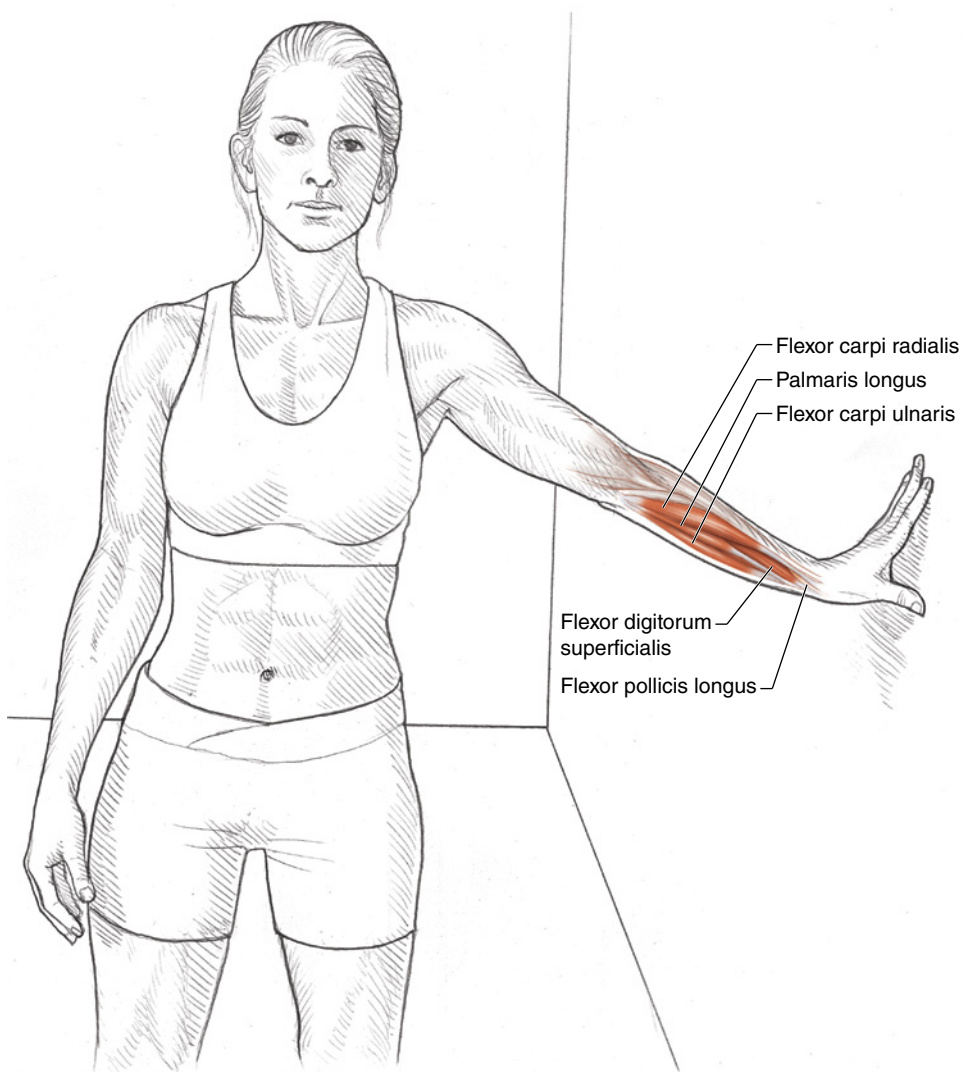
**Less-stretched muscle:** Left flexor pollicis longus

## Stretch Notes

Tightness and hypertonicity of the finger flexors usually arise from making a fist or curling the wrist into flexion. Sleeping with the hands in this position causes the flexor muscle group to become even tighter and shorter, resulting in impingement and damage to the median nerve within the carpal tunnel. The finger flexors also become tight from repetitive work in which the hand grasps something for a long period of time, such as when hammering or rock climbing. A person can also develop what is called trigger finger by overworking the index finger. Some of the problems of the forearm such as golfer's elbow, or medial epicondylitis, are the result of tight finger flexor muscles. Finally, improper hand position in piano playing—wrist not relaxed, using a pushing action rather than a freely rebounding gravity stroke for the key stroke—can lead to finger flexor stiffness.

The elbow angle does not need to be precisely 90 degrees. Choose a comfortable angle. Some people find that fully flexing the elbow makes it easier to push on the hand. With the elbow fully flexed, the push is more down than across.

## WALL-ASSISTED FINGER FLEXOR STRETCH



## Execution

1. Stand upright about 1 foot (30 cm) from a wall.
2. Turn the body so that the left shoulder is perpendicular to the wall.
3. Reach out and place the left fingertips on the wall midway between the left hip and left shoulder.
4. While keeping only the left fingertips in contact with the wall, lean toward the wall.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left flexor carpi radialis, left flexor carpi ulnaris, left flexor digiti minimi brevis, left flexor digitorum profundus, left flexor digitorum superficialis, left palmaris longus

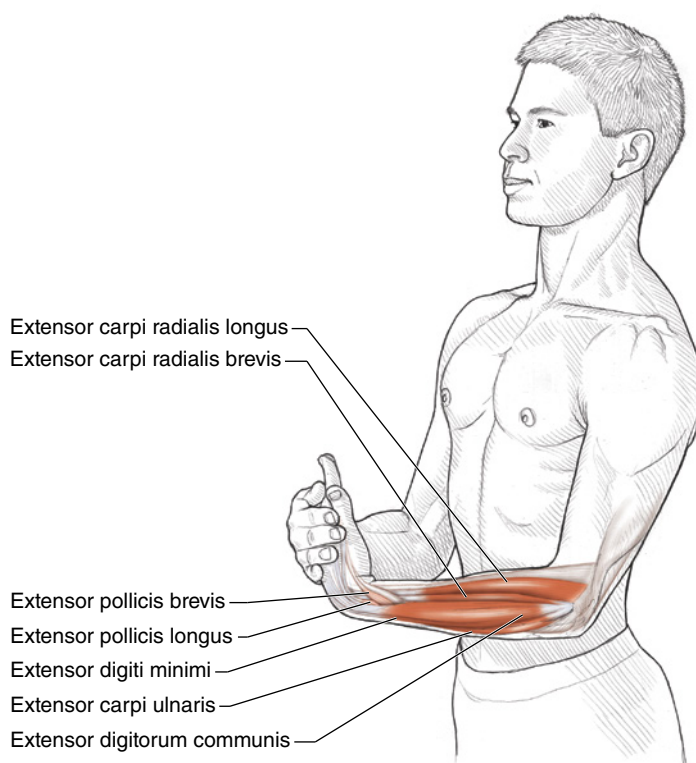
**Less-stretched muscle:** Left flexor pollicis longus

## Stretch Notes

Tightness and hypertonicity of the finger flexors usually arise from making a fist or curling the wrist into flexion. Sleeping with the hands in this position causes the flexor muscle group to become even tighter and shorter, resulting in impingement and damage to the median nerve within the carpal tunnel. The finger flexors also become tight from repetitive work in which the hand grasps something for a long period of time, such as when hammering or rock climbing. A person can also develop what is called trigger finger by overworking the index finger. Some of the problems of the forearm such as golfer's elbow, or medial epicondylitis, are the result of tight finger flexor muscles. Finally, improper hand position in piano playing—wrist not relaxed, with a pushing action rather than a freely rebounding gravity stroke as the key stroke—can lead to finger flexor stiffness.

The initial starting height of your fingers relative to the hips is not critical. You should start in a position that makes it easy to maintain balance while still putting stretch tension on the muscles. As you adapt to the stretch, you may find it necessary to change the finger height to achieve the desired stretching tension.

# FINGER EXTENSOR STRETCH



## Execution

1. Sit or stand upright.
2. Turn the left arm so that the left palm faces up. Flex the left elbow to a 90-degree angle.
3. Flex the left wrist to a 90-degree angle. Flex the fingers so they point toward the elbow.
4. Place the right hand on top of the fingers, and press the fingers down toward the forearm.
5. Repeat these steps for the opposite arm.

## Muscles Stretched

**Most-stretched muscles:** Left extensor carpi radialis brevis, left extensor carpi radialis longus, left extensor carpi ulnaris, left extensor digitorum communis, left extensor digiti minimi, left extensor indicis

**Less-stretched muscles:** Left extensor pollicis brevis, left extensor pollicis longus

## Stretch Notes

Tightness in the extensor muscles is also a cause of tennis elbow, pain in the lateral elbow during arm movements. This tightness usually is caused by overworking or straining these muscles. Therefore, any activity that uses these muscles, such as keyboard work, racket sports, rowing, weightlifting, wheelchair sports, and rock climbing, can lead to overwork, hypertonicity, and tightness. Also, overworking the extensor pollicis longus and brevis or the abductor pollicis longus can lead to conditions known as drummer boy palsy (mainly the extensor pollicis longus) and de Quervain syndrome (mainly the extensor pollicis brevis and abductor pollicis longus). A tight extensor carpi radialis longus or extensor carpi radialis brevis can also lead to inflammation of their respective tendons, which can lead to radial wrist pain or intersection syndrome. By doing this stretch, you help reduce the problems that can arise from overworked finger extensors. Finally, ability to actively extend the finger is used as a reliable early predictor of recovery of arm function in stroke patients. Thus, stretching the finger extensor muscles after a stroke helps in the rehabilitation process.

Increase the magnitude of the stretch by flexing the fingers (i.e., make a fist). Also, the elbow angle does not need to be precisely 90 degrees. Choose a comfortable angle. Some people find that fully flexing the elbow makes it easier to push on the hand. With the elbow fully flexed, the push is more down than across.

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